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B.Tech. Degree I & II Semester (Combined) Supplementary Examination June 2017

IT/CS/CE/ME/SE/EC/EE/EB/EI/FT 1102 ENGINEERING PHYSICS
(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain construction and reconstruction of hologram and explain any three applications.
- (b) Briefly explain different types of losses in optical fibers.
- (c) What are different types of liquid crystals and explain the working of LCD.
- (d) Derive relation between density and lattice constant of a crystal.
- (e) Explain optical, electrical, magnetic and mechanical properties of Nano material.
- (f) Explain length contraction, time dilation and relativistic mass.
- (g) What are matter waves? State and explain uncertainty principle.
- (h) Find resultant sound intensity level, if 60 dB of sound is added to 80 dB of sound.

PART B

(4 × 15 = 60)

- II. (a) Describe the basic components of laser. (4)
- (b) Explain the working of Nd-YAG laser, with the help of a neat diagram. (7)
- (c) Compare hologram with a photograph. (4)
- OR**
- III. (a) What is numerical aperture? Derive its expression. (7)
- (b) Briefly explain pulse dispersion in an optical fiber. (4)
- (c) Briefly explain fibre optic communication system. (4)
- IV. (a) State and derive Bragg's law. (4)
- (b) Explain working of Bragg's X-ray spectrometer. (7)
- (c) Write a short note on shape memory alloys. (4)
- OR**
- V. (a) What are Miller indices? How are they determined? (4)
- (b) Explain properties of metallic glasses. (4)
- (c) What are lattice planes? Derive the expression for spacing between lattice. (7)
- VI. (a) Derive the time dependent Schrodinger equation. (7)
- (b) Explain production of ultrasonic waves, with a neat diagram. (6)
- (c) What accelerating voltage is required to produce an electron of wavelength 0.1 m? (2)
- OR**
- VII. (a) Derive Sabine's formula reverberation. (7)
- (b) Derive energy states and wave function for a particle in a box. (8)
- VIII. (a) Explain the terms: (i) Nano ring. (ii) Nano rod. (iii) Nano shell. (3)
- (b) Explain Michelson Morley experiment in relativity. (9)
- (c) Write note on high temperature super conductivity. (3)
- OR**
- IX. (a) Explain B C S theorem of superconductivity. (6)
- (b) At what speed mass of a body is 20% greater than its rest mass? (3)
- (c) What is Josephson effect? Explain two applications of superconductivity. (6)

